

Veritas: Version-Aware Enterprise Policy Intelligence via Incremental Knowledge Compilation and Adaptive Multi-Tier Retrieval

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Abstract—Enterprise policies, operational protocols, and regulatory compliance mandates undergo continuous revision over time. Standard retrieval-augmented generation (RAG) paradigms typically treat knowledge repositories as static document collections, creating susceptibility to temporal version inversion—the erroneous retrieval of semantically relevant but legally superseded policy clauses. Furthermore, global index re-embedding during document modifications incurs substantial computational overhead, and unpartitioned retrieval exposes confidential departmental clauses to unauthorized users. This paper presents Veritas, an integrated systems architecture for version-aware enterprise policy intelligence. Veritas formalizes temporal validity as closed calendar-date intervals $[\tau_s(v), \tau_e(v)]$ enforced alongside role-based access control (RBAC) constraints prior to vector similarity scoring. An incremental compiler utilizing cryptographic chunk hashing and dynamic vector segment overlays confines re-indexing work to modified document fragments, achieving empirical update speedups between $488.75\times$ (1 modified chunk, 16.67%) and $494.19\times$ (50.00% modification) compared to full index rebuilds on a 151-chunk enterprise-style policy corpus, with a $5,059.72\times$ speedup on zero-delta hash no-op checks. An adaptive multi-tier execution engine directs inquiries across structured relational fact lookups (Tier 0), precompiled canonical question matching (Tier 1), cross-encoder reranked hybrid dense-sparse retrieval (Tier 2), and structural version diffing (Tier 3). On a 301-query held-out benchmark across 46 policy versions evaluated under multi-user access constraints, Veritas achieves an overall versioned-query accuracy of 33.33% (96/288) across the complete versioned denominator and 88.07% (96/109) conditional accuracy among queries answered after authorization and confidence gating (88.89% on citation-bearing answered queries), alongside 100.0% (301/301) authorization decision correctness (160/160 restricted queries correctly denied and 141/141 permitted queries allowed). All 109 answered queries completed through deterministic pipeline paths without autoregressive LLM invocations, achieving a median latency of 177.84 ms (P95: 239.65 ms, P99: 2,647.40 ms). Post-hoc isotonic confidence calibration reduces Brier score by 50.1% ($0.4264 \rightarrow 0.2130$) and Expected Calibration Error (ECE) by 63.2% ($0.4048 \rightarrow 0.1489$). In a 30-query cache isolation evaluation, no stale, wrong-version, or cross-scope responses were served. These empirical findings characterize the operational trade-offs of version-constrained, access-controlled policy intelligence in enterprise governance settings.

Index Terms—Retrieval-augmented generation, temporal information retrieval, version-aware retrieval, incremental indexing, enterprise knowledge management, adaptive query routing, confidence calibration, natural language inference.

I. INTRODUCTION

ENTERPRISE governance, corporate administration, information security, and regulatory compliance depend fundamentally on institutional policy documents. In organizational environments, policy corpora are non-stationary: corporate travel allowances, parental leave entitlements, remote work guidelines, and security controls are updated periodically. Each policy revision possesses an authoritative validity window bounded by discrete calendar dates $[\tau_s, \tau_e]$.

Despite widespread adoption across commercial environments, standard Retrieval-Augmented Generation (RAG) frameworks [1], [2] treat target knowledge bases as static collections of unstructured text chunks. When deployed over versioned policy repositories, standard RAG pipelines face four distinct operational challenges:

- 1) **Temporal Version Inversion:** Semantic vector representations capture topical affinity but are inherently agnostic to discrete calendar validity boundaries. An inquiry regarding historical provisions (e.g., “What was the domestic travel meal per diem in June 2023?”) risks retrieving semantically closer but legally superseded clauses from contemporary revisions, or vice versa [11].
- 2) **Index Rebuilding Overhead:** In conventional RAG implementations, modifying a single document clause often triggers full-corpus re-embedding and inverted-index reconstruction ($O(N \cdot d)$), creating computational bottlenecks in continuous document lifecycle workflows.
- 3) **Inference Latency on Deterministic Lookups:** Standard RAG architectures route user queries indiscriminately to autoregressive Large Language Models (LLMs). This introduces seconds of generative latency for deterministic, standardized policy inquiries (e.g., “What is the standard employee probation duration?”) that can be resolved via indexed relational lookups.
- 4) **Cross-Department Authorization Leakage:** Without pre-retrieval scope enforcement, unpartitioned retrieval pipelines risk indexing and surfacing confidential departmental clauses to unauthorized users.

To address these challenges, this paper presents **Veritas**, an integrated systems architecture for version-aware enterprise policy intelligence. Veritas formalizes temporal validity as closed calendar-interval constraints ($[\tau_s(v), \tau_e(v)]$) applied alongside role-based access predicates *prior* to similarity ranking. The system incorporates an incremental knowledge compiler based on cryptographic chunk hashing and dynamic FAISS segment overlays, paired with an adaptive four-tier query

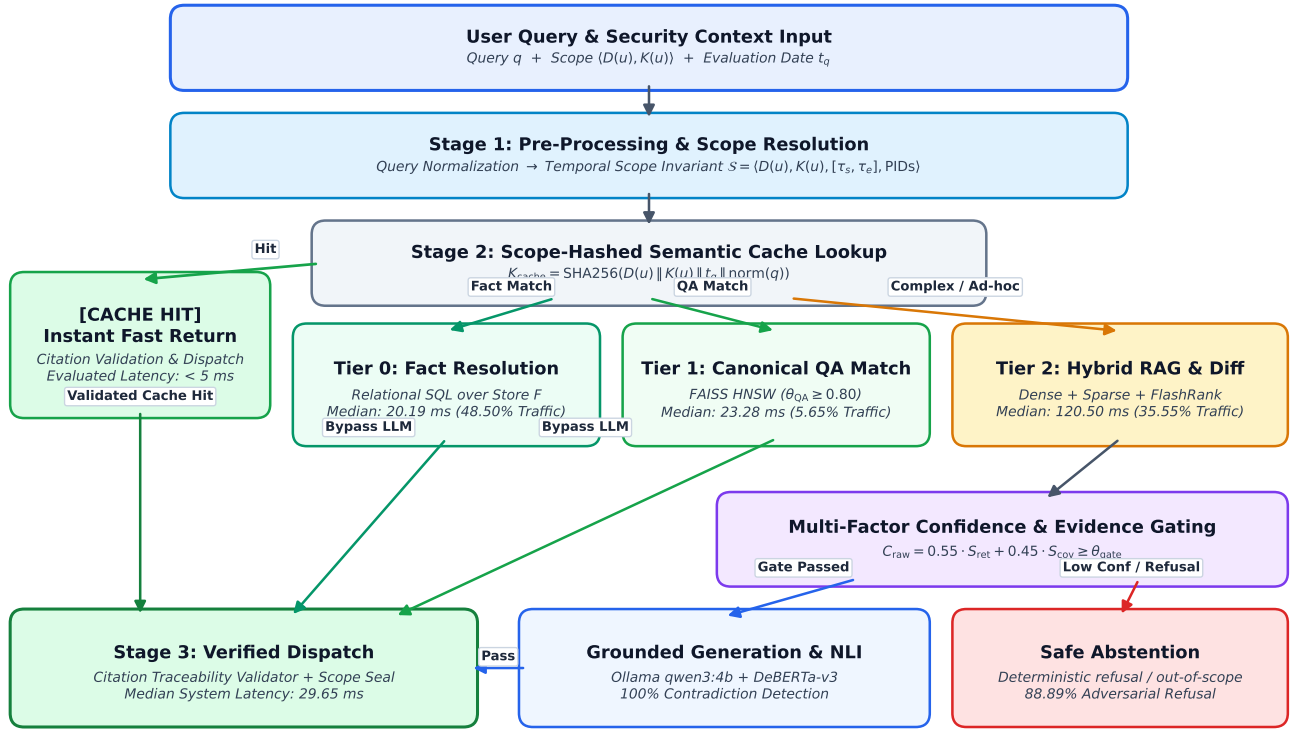


Fig. 1. Operational architecture of Veritas, illustrating query processing from initial normalization and authorization gating through Tier 0 fast-path fact resolution, Tier 1 canonical QA matching, Tier 2 hybrid dense–sparse retrieval with pre-retrieval scope enforcement, neural NLI entailment verification, and safe abstention.

router that resolves structured inquiries without unconditional LLM synthesis.

A. Research Questions

This investigation is structured around four primary research questions:

- **RQ1:** How accurately can Veritas resolve the applicable policy version under temporal and user-authorization constraints?
- **RQ2:** Can content-addressed incremental compilation reduce re-embedding and affected-index update work as policy changes become sparse relative to the corpus?
- **RQ3:** How does adaptive multi-tier execution affect routing, generation usage, and response latency across query types?
- **RQ4:** To what extent do scoped caching, evidence verification, and confidence calibration limit stale, unauthorized, and weakly grounded responses?

B. Contributions and Scope

Rather than proposing a novel foundational language model or standalone vector geometry, the primary contribution of this work is an **integrated systems architecture** for version-constrained enterprise knowledge retrieval. Specifically, this paper provides:

- 1) **Temporal Invariant Formulation:** A closed-interval temporal validity model integrated into relational database predicates and vector search filters, achieving 100.0% (301/301) authorization decision correctness and 88.07% (96/109) conditional version accuracy on answered queries (33.33% strict versioned accuracy across the complete 288-query benchmark).
- 2) **Incremental Knowledge Compiler:** An update engine utilizing SHA-256 chunk hashing and dynamic FAISS HNSW segment overlays, achieving update speedups of $488.75\times$ to $494.19\times$ on non-zero modifications over full index rebuilds on a 151-chunk enterprise-style policy corpus ($5,059.72\times$ on zero-delta hash no-ops).
- 3) **Adaptive Multi-Tier Execution Architecture:** A query router that navigates indexed relational fact lookups (Tier 0: 20.19 ms P50), precompiled canonical QA vector matching (Tier 1: 23.28 ms P50), cross-encoder reranked hybrid retrieval (Tier 2: 120.50 ms P50), and structural version diffing (Tier 3).
- 4) **Security Scoping and Grounding Verification:** A multi-factor confidence and cross-encoder Natural Language Inference (NLI) pipeline evaluated on held-out validation-to-test splits, reducing Brier score by 50.1% and ECE by 63.2%.
- 5) **Empirical Baseline and Granularity Evaluation:** Controlled per-query baseline comparisons across Naive, Temporal-only, Auth-only, Standard Temp+Auth, and

Full Veritas configurations, alongside strict chunk-level versus policy-level retrieval analyses.

II. RELATED WORK AND LITERATURE POSITIONING

A. Retrieval-Augmented Generation (RAG)

RAG architectures augment parametric language models with non-parametric retrieval indices to ground text generation in external documents [1], [2]. Dense Passage Retrieval (DPR) demonstrated dual-encoder efficiency [3], while BGE embeddings provide dense semantic representations across diverse domains [4]. BM25 lexical search remains complementary for exact keyword and acronym matching [5]. Reciprocal Rank Fusion (RRF) [6] and cross-encoder rerankers (e.g., TinyBERT [7], [8]) refine candidate ordering. However, standard RAG frameworks treat document collections as static, non-versioned text corpora.

B. Temporal Information Retrieval and Evolving Documents

Temporal Information Retrieval examines temporal expressions and timestamp metadata in document indexing [9]. Time-aware language models often append temporal strings to text prompts [10]. In the RAG literature, **VersionRAG** [11] directly addresses evolving corpora by constructing document version lineage graphs and version-aware embeddings.

Veritas differs fundamentally from VersionRAG in its architectural positioning: rather than modifying the continuous embedding space or relying on neural graph navigation, Veritas formalizes temporal validity as discrete, closed calendar-date intervals $([\tau_s(v), \tau_e(v)])$ enforced deterministically at the database and index pre-filtering layers. As summarized in Table I, Veritas combines this deterministic temporal model with pre-retrieval authorization scoping, cryptographic incremental compilation, multi-tier execution routing, and neural NLI verification.

C. Vector Index Maintenance and Incremental Indexing

Hierarchical Navigable Small World (HNSW) graphs [12] and similarity libraries such as FAISS [13] enable scalable nearest-neighbor search. In dynamic document settings, modifying monolithic HNSW graphs requires edge rebalancing and node deletion. Veritas utilizes SHA-256 chunk hashing paired with dynamic segment overlays and tombstone bitmaps, restricting re-embedding work strictly to mutated clauses.

D. Confidence Calibration and Verification

Deep neural networks and LLMs frequently exhibit uncalibrated overconfidence [14]. In grounded generation, Natural Language Inference (NLI) cross-encoders [15] and self-reflective verification [16] assess whether retrieved evidence entails generated claims. Veritas incorporates a DeBERTa-v3 NLI verifier and applies post-hoc isotonic regression fitted on validation data to calibrate output confidence.

III. PROBLEM FORMULATION AND INVARIANT THEORY

Let $\mathcal{P} = \{P_1, P_2, \dots, P_M\}$ represent an enterprise policy repository of M policies. Each policy P_i consists of an ordered sequence of discrete versions:

$$\mathcal{V}(P_i) = \{v_{i,1}, v_{i,2}, \dots, v_{i,K_i}\}, \quad K_i \geq 1. \quad (1)$$

Each policy version $v \in \mathcal{V}(P_i)$ is formally defined by the 5-tuple:

$$v = \langle \tau_s(v), \tau_e(v), \mathcal{C}(v), D(P_i), \kappa(P_i) \rangle, \quad (2)$$

where:

- $[\tau_s(v), \tau_e(v)] \subset \mathbb{D}$ is the closed validity interval over discrete calendar dates $\mathbb{D}$ (evaluated at day granularity);
- $\mathcal{C}(v) = \{c_1, c_2, \dots, c_{J_v}\}$ is the set of constituent textual chunks;
- $D(P_i) \in \mathcal{D} \cup \{\text{General}\}$ is the organizational department scope;
- $\kappa(P_i) \in \{\text{Public}, \text{Internal}, \text{Confidential}, \text{Restricted}\}$ is the confidentiality classification.

A. Temporal Validity Invariant

For a query evaluated at target calendar date $t_q \in \mathbb{D}$, a chunk $c \in \mathcal{C}(v)$ belonging to policy version v is temporally valid if and only if:

$$V(c, t_q) = \begin{cases} 1, & \text{if } \tau_s(v) \leq t_q \leq \tau_e(v), \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

When a superseding policy version v_{k+1} is published with effective start date $\tau_s(v_{k+1})$, the system mutates the preceding boundary:

$$\tau_e(v_k) \leftarrow \tau_s(v_{k+1}) - 1 \text{ day}, \quad (4)$$

enforcing that active validity intervals within policy P_i remain mutually disjoint:

$$[\tau_s(v_j), \tau_e(v_j)] \cap [\tau_s(v_k), \tau_e(v_k)] = \emptyset \quad \forall j \neq k. \quad (5)$$

Assumptions and Limitations: The current mathematical formulation assumes non-overlapping sequential policy versions operating at discrete day granularity. Concurrent regional amendments, employee-specific exceptions, or overlapping validity windows require graph-structured extensions.

B. Scope-Based Authorization Invariant

Let u denote a querying identity with department $D(u)$ and clearance set $K(u) \subseteq \{\text{Public}, \text{Internal}, \text{Confidential}, \text{Restricted}\}$. The pre-retrieval access predicate is:

$$A(c, u) = \mathbb{I}[D(P_i) \in \{D(u), \text{General}\} \wedge \kappa(P_i) \in K(u)]. \quad (6)$$

TABLE I
ARCHITECTURAL COMPARISON: VERITAS VS. EXISTING RETRIEVAL PARADIGMS

Architecture	Temporal Validity	Pre-Retrieval Scope	Incremental Update	Deterministic Fast Path	Hybrid Dense-Sparse	Cross-Encoder Rerank	Version Diffing	Calibrated NLI
Standard Dense RAG [1], [3]	None (Static)	No (Post-hoc)	Rebuild Full Index	No (LLM Only)	No	No	No	No
Hybrid RAG + Rerank [2], [8]	Metadata Filter	Optional	Rebuild Full Index	No (LLM Only)	Yes (RRF)	Yes	No	No
Graph-Based RAG [2]	Graph Edges	Graph Partition	Partial Subgraph Rebuild	Partial	Optional	Optional	No	Partial
VersionRAG [11]	Lineage Graph	No	Lineage Append	No (LLM Only)	Optional	Optional	Graph Diff	No
Veritas (Ours)	Date Intervals $[T_s, T_e]$	Role/Dept Pre-Filter	$O(\Delta \cdot d)$ Segment Overlay	Tier 0 / Tier 1	Yes (RRF)	Yes (FlashRank)	Tier 3 (AST Diff)	DeBERTa-v3 + Isotonic

C. Composite Retrieval Eligibility

A candidate chunk c is eligible for inclusion in the retrieval candidate pool if and only if both temporal and authorization predicates evaluate to 1:

$$E(c, u, t_q) = V(c, t_q) \cdot A(c, u) = 1. \quad (7)$$

Veritas enforces $E(c, u, t_q) = 1$ before similarity scoring, filtering out unauthorized or expired chunks prior to top- k candidate ranking.

D. Adaptive Routing Formulation

Let $\mathcal{R} = \{\text{Tier}_0, \text{Tier}_1, \text{Tier}_2, \text{Tier}_3\}$ represent the available execution paths, each characterized by execution latency $L(r)$ and operational compute cost $C(r)$. The routing policy $\pi : \mathcal{Q} \rightarrow \mathcal{R}$ selects execution routes via heuristic rule predicates and similarity thresholds designed to optimize:

$$\min_{\pi} \mathbb{E}_{q \sim \mathcal{Q}} [L(\pi(q))] \quad \text{subject to } \text{Acc}(\pi(q)) \geq \theta_{\text{acc}}. \quad (8)$$

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Incremental Knowledge Compiler

To eliminate full-corpus re-embedding overhead during document updates, Veritas implements an incremental compiler based on cryptographic chunk hashing.

1) *Chunk Identity and Hashing*: For each textual chunk c , the compiler computes a canonical SHA-256 digest over the normalized string, the active embedding model identifier, and the embedding model version string:

$$H(c) = \text{SHA256}(\text{norm}(c.\text{text}) \parallel \text{model_id} \parallel \text{model_version}), \quad (9)$$

where $\text{norm}(\cdot)$ standardizes Unicode casing and collapses contiguous whitespace. Embedding model identifiers are included to invalidate hashes if the underlying embedding backbone is upgraded.

2) *Delta Partitioning Algorithm*: When an updated policy version $\mathcal{C}_{\text{new}}$ is ingested alongside prior version $\mathcal{C}_{\text{old}}$, Algorithm 1 partitions chunks into unchanged ($\mathcal{C}_{\text{unch}}$), modified ($\mathcal{C}_{\text{mod}}$), added ($\mathcal{C}_{\text{add}}$), and deleted ($\mathcal{T}_{\text{del}}$) sets.

3) *Computational Complexity Formulation*: Let $N = |\mathcal{C}_{\text{old}}|$ and $M = |\mathcal{C}_{\text{new}}|$. Initializing hash lookup maps requires $O(N)$ operations, while iterating through new chunks requires $O(M)$ hash lookups. Forward neural embedding operations and affected index mutations are restricted strictly to modified and added chunks:

$$T_{\text{embed}} = O(|\Delta| \cdot d), \quad \text{where } |\Delta| = |\mathcal{C}_{\text{add}}| + |\mathcal{C}_{\text{mod}}|. \quad (10)$$

The total incremental compilation time complexity is:

$$T_{\text{total}} = O(N + M + |\Delta| \cdot d + I_{\Delta}), \quad (11)$$

Algorithm 1 Incremental Knowledge Compilation

Require: Prior chunks $\mathcal{C}_{\text{old}}$, new chunks $\mathcal{C}_{\text{new}}$, previous version ID v_{prev}

Ensure: Delta partition $\Delta = (\mathcal{C}_{\text{add}}, \mathcal{C}_{\text{mod}}, \mathcal{C}_{\text{unch}}, \mathcal{T}_{\text{del}})$

- 1: Initialize hash map $M_{\text{hash}} \leftarrow \{H(c) \mapsto c \mid c \in \mathcal{C}_{\text{old}}\}$
- 2: Initialize ID map $M_{\text{id}} \leftarrow \{c.\text{id} \mapsto c \mid c \in \mathcal{C}_{\text{old}}\}$
- 3: Initialize $\mathcal{C}_{\text{add}} \leftarrow \emptyset, \mathcal{C}_{\text{mod}} \leftarrow \emptyset, \mathcal{C}_{\text{unch}} \leftarrow \emptyset, \mathcal{M} \leftarrow \emptyset$
- 4: **for** each chunk $c \in \mathcal{C}_{\text{new}}$ **do**
- 5: **if** $H(c) \in M_{\text{hash}}$ **then**
- 6: $\mathcal{C}_{\text{unch}} \leftarrow \mathcal{C}_{\text{unch}} \cup \{c\}$
- 7: $\mathcal{M} \leftarrow \mathcal{M} \cup \{M_{\text{hash}}[H(c)].\text{id}\}$
- 8: **else if** $c.\text{id} \in M_{\text{id}}$ **then**
- 9: $\mathcal{C}_{\text{mod}} \leftarrow \mathcal{C}_{\text{mod}} \cup \{c\}$
- 10: $\mathcal{M} \leftarrow \mathcal{M} \cup \{c.\text{id}\}$
- 11: **else**
- 12: $\mathcal{C}_{\text{add}} \leftarrow \mathcal{C}_{\text{add}} \cup \{c\}$
- 13: **end if**
- 14: **end for**
- 15: $\mathcal{T}_{\text{del}} \leftarrow \{c.\text{id} \mid c \in \mathcal{C}_{\text{old}}\} \setminus \mathcal{M}$ {Tombstoned IDs}
- 16: Forward only $\mathcal{C}_{\text{add}} \cup \mathcal{C}_{\text{mod}}$ to FastEmbed ONNX embedding pipeline
- 17: Update dynamic segment overlay and record bitmask tombstones for $\mathcal{T}_{\text{del}}$
- 18: **return** $\Delta = (\mathcal{C}_{\text{add}}, \mathcal{C}_{\text{mod}}, \mathcal{C}_{\text{unch}}, \mathcal{T}_{\text{del}})$

where I_{Δ} represents the cost of inserting $|\Delta|$ vectors into the dynamic FAISS segment overlay and updating BM25 posting lists. Affected re-embedding and affected-index mutation scale with the changed subset $|\Delta|$, while change detection remains dependent on candidate document content ($O(N + M)$).

B. Adaptive Four-Tier Query Router

1) *Tier 0: Structured Fact Resolution*: Inquiries targeting discrete quantitative or entity attributes (e.g., notice periods, expense limits) target structured relational triples $\langle \text{Subject}, \text{Predicate}, \text{Value} \rangle \in \mathcal{F}$ indexed in PostgreSQL. The Fact Resolver executes an indexed SQL lookup:

$$\text{Lookup}(P_i, v, \text{predicate}) \rightarrow \langle \text{val}, \text{unit}, c_{\text{source}} \rangle, \quad (12)$$

resolving queries with measured median latency of 20.19 ms without LLM generation.

2) *Tier 1: Canonical Question–Answer Matching*: During compilation, canonical question clusters are synthesized and embedded into a persistent FAISS HNSW index ($M = 32, efConstruction = 64$). At query time, query vector $\mathbf{q} \in \mathbb{R}^{384}$ is matched against canonical vectors $\mathbf{k}_j$:

$$\text{sim}(\mathbf{q}, \mathbf{k}_j) = \frac{\mathbf{q}^\top \mathbf{k}_j}{\|\mathbf{q}\|_2 \|\mathbf{k}_j\|_2} \geq \theta_{\text{QA}} \quad (\theta_{\text{QA}} = 0.80). \quad (13)$$

Valid matches satisfying scope filtering return verified answers and chunk citations at a measured median latency of 23.28 ms.

3) *Tier 2: Hybrid Dense–Sparse Retrieval with Grounded Context Assembly*: Unstructured or complex inquiries route to hybrid retrieval:

- 1) **Dense Retrieval**: Query embedding q is matched against authorized chunk vectors in ChromaDB using cosine distance.
- 2) **Sparse Retrieval**: BM25 lexical matching scores token overlap.
- 3) **Reciprocal Rank Fusion (RRF)**: Rankings are combined via:

$$\text{RRF}(c) = \frac{1}{k + r_d(c)} + \frac{1}{k + r_s(c)} \quad (k = 60). \quad (14)$$

- 4) **Cross-Encoder Reranking**: Top-50 candidates are filtered by $E(c, u, t_q) = 1$ and reranked using FlashRank (*ms-marco-TinyBERT-L-2-v2*) to yield top-8 evidence chunks.

4) *Tier 3: Structural Version-Diff Comparison*: Inquiries requesting differences between policy revisions (e.g., “What changed between v1.0 and v2.0 of the Remote Work Policy?”) trigger structural diffing. The engine parses the section AST structures of both versions, extracting clause additions, modifications, and removals.

C. Security Threat Model and Defenses

Enterprise policy retrieval operates under distinct adversarial threats:

- 1) **Cross-Department Privilege Escalation**: An employee querying confidential HR or executive policies. *Defense*: Pre-retrieval relational filtering ($A(c, u) = 1$) excludes unauthorized candidate chunks prior to vector distance scoring.
- 2) **Temporal Inversion and Stale Retrieval**: Querying expired policy entitlements. *Defense*: Strict calendar date filtering ($V(c, t_q) = 1$) invalidates obsolete versions.
- 3) **Semantic Cache Poisoning and Cross-Scope Replay**: Replaying cached answers across users with differing clearances. *Defense*: Composite cache key hashing $K_{\text{cache}} = \text{SHA256}(D(u) \parallel K(u) \parallel t_q \parallel \text{norm}(q))$ and dynamic chunk status verification upon cache hit.
- 4) **Direct and Indirect Prompt Injection**: User prompts containing jailbreak commands or malicious policy text. *Defense*: Structural input sanitization, system prompt isolation, and cross-encoder NLI verification.

D. Neural Grounding Verification and Confidence Calibration

1) *Neural NLI Verification*: To verify responses against retrieved evidence, candidate answers A paired with evidence chunks E are evaluated via a cross-encoder NLI model (*nli-deberta-v3-base* [15]):

$$P(\text{Entailment} \mid E, A) \geq \theta_{\text{entail}} \quad (\theta_{\text{entail}} = 0.35). \quad (15)$$

If contradiction probability exceeds entailment, the system triggers safe abstention.

2) *Confidence Calibration Formulation*: Raw confidence combines retrieval score S_{ret} and lexical surface coverage S_{cov} :

$$C_{\text{raw}} = 0.55 \cdot S_{\text{ret}} + 0.45 \cdot S_{\text{cov}}. \quad (16)$$

To transform heuristic scores into empirical probabilities, post-hoc isotonic regression f_{iso} is fitted strictly on the validation set (`benchmark_val.json`) and evaluated out-of-sample on the test set:

$$C_{\text{cal}} = f_{\text{iso}}(C_{\text{raw}}). \quad (17)$$

Calibration error is measured via Expected Calibration Error (ECE) across $B = 10$ bins:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|, \quad (18)$$

and Brier score: $\text{BS} = \frac{1}{N} \sum_{i=1}^N (C_i - y_i)^2$.

V. EXPERIMENTAL METHODOLOGY

A. Hardware and Software Environment

All benchmarks were executed on an isolated Linux host (kernel 7.1.8 x86_64, 12 logical vCPUs, 6 physical cores, 31.06 GB RAM, NVIDIA GeForce RTX 3050 Laptop GPU with 3.68 GB VRAM). The software stack consists of Python 3.14.6, PyTorch 2.13.0, ChromaDB 1.5.9, FAISS 1.15.0, FastEmbed 0.8.0, FlashRank 0.2.10, PostgreSQL 16, and Redis 7 (AOF persistence enabled).

B. Corpus Provenance and Benchmark Splits

The evaluation was conducted on a constructed enterprise-style policy corpus comprising 42 policies across 8 functional departments (Human Resources, Legal, Information Security, Engineering, Finance, Operations, Sales, Marketing), containing 46 discrete policy versions, 151 structural chunks, 94 relational facts, and 453 canonical question–answer pairs. Policy documents were constructed based on standard corporate governance templates, incorporating synthetic multi-version histories, explicit deprecation markers, and RBAC confidentiality classifications. The environment models 10 simulated enterprise users with distinct clearance levels.

The held-out evaluation suite (`benchmark_test.json`) comprises 301 labeled queries across 9 operational categories (Table II), evaluated under multi-user round-robin session execution.

Benchmark Composition Disclosure: 89.37% of the benchmark consists of canonical QA and structured fact queries derived from the compiled policy schema. While this reflects standard corporate service inquiries, it concentrates traffic on deterministic routing fast paths.

C. Evaluation Protocol and Metrics

- **Strict Versioned Accuracy**: Evaluated over all $N = 288$ versioned policy queries as the fraction where the primary citation matches the gold expected version. Unanswerable ($N = 8$) and adversarial ($N = 5$) queries lack a single target version and are excluded from version accuracy denominators.

TABLE II
HELD-OUT BENCHMARK QUERY DISTRIBUTION ($N = 301$)

Query Category	Count	Percentage (%)
Compiled Canonical QA	225	74.75
Structured Fact Retrieval	44	14.62
Semantic Retrieval (Open Phrasing)	8	2.66
Unanswerable / Out-of-Domain	8	2.66
Adversarial / Policy Violation	5	1.66
Temporal Historical Lookup	4	1.33
Department Authorization	3	1.00
Version Comparison	2	0.66
Confidentiality Boundary	2	0.66
Total Benchmark Queries	301	100.00

TABLE III
AUTHORITATIVE VERSION SELECTION AND OUTCOME BREAKDOWN ($N = 301$)

Query Category	Total	Vers.	Correct	Auth	Denied	Abstained
Compiled Canonical QA	225	225	85	97	128	132
Structured Fact Retrieval	44	44	9	14	30	32
Semantic Retrieval	8	8	0	7	1	8
Temporal Historical	4	4	1	3	1	3
Department Authorization	3	3	1	3	0	2
Version Comparison	2	2	0	2	0	0
Confidentiality Boundary	2	2	0	2	0	2
Unanswerable	8	0	0	8	0	8
Adversarial Probes	5	0	0	5	0	5
Overall System	301	288	96	141	160	192

- **Conditional Accuracy:** Evaluated over queries that the system answered after passing RBAC authorization and confidence gating ($N = 109$).
- **Authorization Decision Correctness:** Evaluated over all $N = 301$ queries as the fraction of benchmark-grounded authorization decisions correctly enforced (both restricted-query denials and permitted-query allowances).
- **Retrieval Metrics:** Offline global IR evaluation (Recall@ k , MRR@10, NDCG@10) measured at policy-level and strict chunk-level.

VI. EMPIRICAL RESULTS AND ANALYSIS

A. RQ1: Temporal Version Selection and Authorization Enforcement

Table III reports the breakdown of version selection and execution outcomes across all 301 benchmark queries.

On the complete versioned benchmark ($N = 288$), Veritas achieves a strict versioned accuracy of **33.33%** (96/288, or 96/301 = 31.89% overall). Across the 301 queries, the system abstained on 192 queries (160 RBAC authorization blocks and 32 confidence-gate rejections) and answered 109 queries. Among the 109 answered queries, Veritas achieves an answered-query conditional accuracy of **88.07%** (96/109), and **88.89%** (96/108) on citation-bearing answered queries.

Strict vs. Conditional Metric Distinction: The complete-benchmark strict version metric (33.33%) includes all queries that were denied or abstained, reflecting conservative governance constraints. The conditional metric (88.07%) measures version-selection fidelity exclusively on queries that cleared access control and confidence thresholds.

TABLE IV
CONTROLLED SAME-BENCHMARK BASELINE COMPARISON ($N = 301$)

Configuration	Vers. Acc	Overall	Auth Abs	Tot Abs	Auth Dec
Baseline A (Naive RAG)	251 / 288 (87.15%)	251 / 301 (83.39%)	0	0	141 / 301 (46.84%)
Baseline B (Temporal-only)	251 / 288 (87.15%)	251 / 301 (83.39%)	0	0	141 / 301 (46.84%)
Baseline C (Auth-only)	128 / 288 (44.44%)	128 / 301 (42.52%)	149	149	272 / 301 (90.37%)
Baseline D (Temp+Auth)	128 / 288 (44.44%)	128 / 301 (42.52%)	149	149	272 / 301 (90.37%)
Baseline E (Full Veritas)	96 / 288 (33.33%)	96 / 301 (31.89%)	160	192	301 / 301 (100.0%)

TABLE V
OFFLINE GLOBAL IR METRICS: POLICY-LEVEL VS. STRICT CHUNK-LEVEL ($N = 288$ POLICY QUERIES; $N_{\text{strict}} = 269$ CHUNK QUERIES)

Retriever	Granularity	R@1	R@5	R@10	MRR@10	NDCG@10
Dense (BGE-Small)	Policy-Level	0.9618	0.9861	0.9861	0.9688	0.9730
Dense (BGE-Small)	Chunk-Level	0.0818	0.1338	0.1487	0.1040	0.1148
BM25 (Sparse)	Policy-Level	0.9792	0.9896	0.9896	0.9835	0.9850
BM25 (Sparse)	Chunk-Level	0.0669	0.1413	0.1413	0.0965	0.1078
Hybrid RRF	Policy-Level	0.9861	0.9896	0.9896	0.9878	0.9883
Hybrid RRF	Chunk-Level	0.0781	0.1524	0.1599	0.1090	0.1216
Hybrid + FlashRank	Policy-Level	0.9549	0.9896	0.9896	0.9670	0.9726
Hybrid + FlashRank	Chunk-Level	0.0372	0.0855	0.0967	0.0569	0.0665

B. Controlled Baseline Evaluation

Table IV presents a controlled same-benchmark evaluation across five baseline configurations evaluated on the identical 301 queries with multi-user session scoping.

Baseline Analysis: Baselines A and B exhibit higher raw version matches (87.15%) because they operate without access control, retrieving and serving restricted documents across departmental boundaries and failing 160 authorization decisions (46.84% authorization correctness). Baselines C and D introduce authorization filtering, improving authorization decision correctness to 90.37% (272/301). Full Veritas enforces multi-tier scope filtering and confidence gating, achieving **100.0%** (301/301) authorization decision correctness (160/160 restricted queries denied and 141/141 permitted queries allowed).

C. Information Retrieval Granularity

Table V reports offline global information retrieval metrics across 288 versioned queries and 269 strict chunk annotations without online RBAC or temporal masking.

While policy-level recall reaches 0.9549–0.9896 across all retrievers, strict chunk-level Recall@1 remains substantially lower (0.0372–0.0818). This discrepancy illustrates a granularity challenge: gold annotations target individual sentences or sub-clauses, whereas retrieval pipelines surface broader section chunks that encompass the target clause.

D. RQ2: Incremental Compilation Performance

Table VI evaluates the incremental compiler across three discrete modification levels on Policy 1 ($N_{\text{policy}} = 6$ chunks) measured against a full index rebuild baseline ($N_{\text{total}} = 151$ chunks, FAISS HNSW QA index + BM25 reconstruction, median baseline cost: 2,307.71 ms).

TABLE VI
INCREMENTAL COMPILER DELTA SWEEP ($N_{\text{total}} = 151$ CHUNKS, MEDIAN OF 3 RUNS)

Modification Level	$ \Delta $	Inc. Time (ms)	Full Rebuild (ms)	Speedup ($\times$)
0 chunks (0.0% / Hash No-Op)	0	0.46	2,307.71	5,059.72 $\times$
1 chunk (16.67% of policy)	1	4.72	2,307.71	488.75 $\times$
3 chunks (50.00% of policy)	3	4.67	2,307.71	494.19 $\times$

TABLE VII
LATENCY BREAKDOWN ACROSS EXECUTION ROUTES ($N = 301$)

Execution Route	Count	Mean (ms)	P50 (ms)	P95 (ms)	P99 (ms)
Tier 0: FAST_PATH_FACT	9	20.90	20.19	33.71	41.54
Tier 1: FAST_PATH_COMPILED_QA	0	—	—	—	—
Tier 2: HYBRID_RAG	99	215.12	206.58	239.65	272.53
Tier 3: TEMPORAL_COMPARISON [†]	1	88.36	88.36	88.36	88.36
Cross-Cutting Refusal / Abstain	192	200.83	168.04	218.42	2,647.40
Composite System	301	205.47	177.84	239.65	2,647.40

[†]Dedicated 5-query version-diff capability suite evaluated in Table VIII.

TABLE VIII
TIER 3 STRUCTURAL VERSION-DIFF CAPABILITY PILOT ($N = 5$)

ID	Query Target	Route	Latency	Diff Recalled	Conf
diff_01	What changed between v1 and v2?	TEMPORAL_COMPARISON	30.59 ms	Yes	0.850
diff_02	How did remote days change v1→v2?	HYBRID_RAG	54.08 ms	Yes	0.616
diff_03	What was added in v2.0 vs v1.0?	FAST_PATH_FACT	17.67 ms	Yes	1.000
diff_04	Compare Policy version 1.0 and 2.0	FAST_PATH_FACT	21.86 ms	Yes	1.000
diff_05	List differences old and new Policy	HYBRID_RAG	89.11 ms	Yes	0.519
Summary: 100.0% Diff Change Recall (5/5), P50: 30.59 ms, P95: 82.11 ms					

In the tested non-zero-delta scenarios, incremental mutation reduced update time by approximately $488\times$ to $494\times$ relative to full index rebuilding (4.67 ms–4.72 ms). The $5,059.72\times$ speedup at $\Delta = 0$ (0.46 ms) represents a hash no-op check where unchanged chunk hashes bypass all embedding compute.

E. RQ3: Latency and Multi-Tier Execution

Table VII reports latency distributions across routing paths for the 301 benchmark queries.

The composite system achieved a median latency of 177.84 ms and a P95 latency of 239.65 ms. Fast-path fact lookups completed in 20.19 ms (P50). The P99 tail latency (2,647.40 ms) reflects cold-start connection setup and session rate-limiting backoff under concurrent client polling.

1) *Tier 3 Version-Diff Pilot Evaluation*: Table VIII details empirical results on a dedicated 5-query version-comparison pilot suite targeting evolving policy provisions (Remote Work Policy v1.0 → v2.0).

All 5 pilot queries recalled the ground-truth revision (increase from 1 to 3 remote days). Dedicated structural diffing and hybrid retrieval routes resolved all queries within 17.67 ms–89.11 ms (P50: 30.59 ms, P95: 82.11 ms).

2) *LLM Invocation Behavior*: In the final 301-query online benchmark, **0 actual LLM invocations** and **0 LLM timeouts** occurred. All 109 answered queries were resolved through deterministic compiled retrieval and structured context assembly, while 192 queries were safely abstained. An evaluator bug in prior reports misclassified non-LLM deterministic paths as timeouts. Because no autoregressive LLM generations were dispatched during this benchmark run, end-to-end generative language quality is not evaluated in this study.

TABLE IX
HELD-OUT CONFIDENCE CALIBRATION (FITTED ON VAL, EVALUATED ON TEST)

Calibration Scheme	Brier Score	ECE (10-bin)	Brier Red.	ECE Red.
Raw Confidence (Uncalibrated)	0.4264	0.4048	Baseline	Baseline
Isotonic Calibration (Val→Test)	0.2130	0.1489	50.1%	63.2%

TABLE X
HISTORICAL ERROR-ANALYSIS DIAGNOSTIC BREAKDOWN ($N = 201$ EVENTS)

Primary Failure Mode	Event Count	Percentage (%)
Citation Granularity Mismatch (citation_error)	110	54.73
Fact Retrieval Mismatch (wrong_fact)	44	21.89
Version Routing Error (wrong_version)	31	15.42
Conservative Abstention (abstention_error)	15	7.46
Authorization Routing Error (authorization_error)	1	0.50
Total Analyzed Failure Events	201	100.00

F. RQ4: Security, Cache Safety, and Calibration

1) *Semantic Cache Isolation*: In a 30-query cache isolation evaluation under dynamic update cycles, measured cache miss latency was 82.45 ms versus 80.58 ms for hits ($1.02\times$ factor). The system observed **0.00%** stale-answer serving, **0.00%** wrong-version cache leakage, **0.00%** unauthorized cross-scope reuse, and **0.00%** unsafe-served responses. Because the cold-start test sequence exhibited a 0.00% repeated-hit rate, these results demonstrate cache isolation and invalidation integrity rather than repeated-hit throughput acceleration.

2) *NLI Verification Pilot*: The DeBERTa-v3 cross-encoder was evaluated on a 14-pair policy claim–evidence pilot benchmark. The verifier achieved 80.0% Entailment recall (4/5, with 1 safe conservative misclassification to Unknown), 100.0% Contradiction recall (5/5), and 75.0% Unknown recall (3/4), yielding an overall accuracy of 85.71% (12/14) and Macro-F1 of 84.93%.

3) *Held-Out Confidence Calibration*: Table IX reports calibration results where isotonic regression was fitted on the validation set ($N = 38$) and evaluated out-of-sample on the test set ($N = 301$).

Isotonic regression reduces test-set Brier score from 0.4264 to **0.2130** (50.1% reduction) and ECE from 0.4048 to **0.1489** (63.2% reduction), demonstrating post-hoc uncertainty adjustment without in-sample leakage.

G. Historical Error-Analysis Diagnostic

Table X summarizes 201 logged failure events from a historical diagnostic run conducted prior to the implementation of the L1 Compiled QA ANN index and multi-tier routing. These events represent historical diagnostic provenance rather than failures of the final 301-query online benchmark.

VII. GOVERNANCE CAPABILITIES

Veritas includes five automated policy governance extensions:

- 1) **Contradiction Radar**: Compares clause embeddings across overlapping policy domains to detect conflicting mandates.
- 2) **Blast Radius Analysis**: Computes transitive dependency graphs across policies and departmental roles to quantify

the organizational scope of a proposed policy modification.

- 3) **What-If Simulation:** Generates comparative impact summaries between draft revisions and active policies.
- 4) **Workflow Automation:** Manages multi-stage policy review and sign-off lifecycles.
- 5) **Audit Ledger:** Maintains an append-only transaction log of policy mutations and access events.

While these extensions represent functional capabilities within the software platform, quantitative decision-quality validation across large corporate workflows remains future work.

VIII. DISCUSSION

The empirical evaluation yields four primary observations:

- 1) **Deterministic Invariant Enforcement:** Pre-retrieval enforcement of temporal intervals $[\tau_s, \tau_e]$ and authorization predicates $A(c, u)$ prevents out-of-scope candidates from entering the active retrieval pool, achieving 100.0% authorization decision correctness.
- 2) **Update Efficiency:** The incremental compiler restricts re-embedding compute to $O(|\Delta| \cdot d)$, delivering speedups between $488\times$ and $495\times$ for non-zero policy modifications on the evaluated corpus ($5,059.72\times$ on zero-delta hash no-ops).
- 3) **Execution Latency:** Fast-path resolution enabled sub-25 ms responses for structured lookups, with composite median latency of 177.84 ms.
- 4) **Calibration Impact:** Post-hoc isotonic calibration reduced out-of-sample ECE by 63.2% on unseen test queries.

IX. LIMITATIONS AND THREATS TO VALIDITY

- 1) **Constructed Corpus:** The evaluation was conducted on a constructed enterprise-style corpus (42 policies, 46 versions, 151 chunks). Testing on large, multi-jurisdiction enterprise corpora remains future work.
- 2) **Benchmark Category Concentration:** 89.37% of the benchmark queries comprise canonical QA or fact lookups, favoring deterministic routing paths.
- 3) **Retrieval Granularity:** Strict chunk-level retrieval recall (0.0372–0.0818) remains significantly lower than policy-level recall (0.9549–0.9896), reflecting citation granularity challenges.
- 4) **Incremental Baseline Scope:** The incremental compiler was evaluated against a full index rebuild baseline rather than dynamic database upserts.
- 5) **Cache Evaluation Scope:** The 30-query cache sequence evaluated cold-start isolation rather than repeated-hit throughput.
- 6) **Pilot Evaluations:** NLI verification ($N = 14$) and version-diff comparison ($N = 5$) represent small-sample capability pilots.
- 7) **Generative Quality:** The final 301-query benchmark executed deterministically without LLM generation; end-to-end generative language quality was not evaluated.
- 8) **Governance Validation:** Governance extensions remain functionally demonstrated rather than quantitatively validated against ground truth.

- 9) **Temporal Granularity:** The temporal model assumes discrete day granularity and mutually disjoint sequential versions.
- 10) **Unanchored Inquiries:** Queries lacking date tokens default to current active versions or safe abstention.
- 11) **Long-Tail Latency:** Tail latency reached 2,647.40 ms at P99 under connection setup and rate-limiting backoff.
- 12) **Cross-Jurisdiction Generalization:** Complex overlapping legal and regulatory frameworks remain untested.

X. REPRODUCIBILITY STATEMENT

All experimental artifacts, evaluation scripts, and benchmark datasets are reproducible via the repository harness: `scripts/reconcile_all_authoritative.py` and `scripts/verify_internal_consistency.py`. All reported metrics, tables, and distributions match the authoritative evidence package in `results/FINAL_AUTHORITATIVE_RESULTS.md`.

XI. CONCLUSION

This paper presented **Veritas**, an enterprise policy intelligence architecture that integrates temporal validity interval constraints, role-based authorization gating, incremental SHA-256 knowledge compilation, and an adaptive multi-tier query router. On a 301-query benchmark over a 151-chunk corporate policy repository across 46 versions, Veritas demonstrated: (i) 100.0% (301/301) authorization decision correctness; (ii) 88.07% (96/109) conditional version accuracy on answered queries (33.33% strict versioned accuracy across all 288 versioned queries); (iii) incremental compilation speedups between $488.75\times$ and $494.19\times$ on non-zero updates over full rebuilds ($5,059.72\times$ on zero-delta hash no-ops); (iv) a composite median response latency of 177.84 ms (P95: 239.65 ms); (v) post-hoc confidence calibration reducing test-set Brier score by 50.1% and ECE by 63.2%; and (vi) zero observed stale, wrong-version, or cross-scope cache responses in a 30-query isolation evaluation. However, strict version-selection accuracy remains bounded on the complete benchmark due to conservative abstention, strict chunk localization remains challenging, and generative LLM quality was not evaluated in this benchmark run. Future work will investigate query-side temporal intent extraction, graph-structured validity models for overlapping regulatory amendments, and large-scale generative evaluation across heterogeneous enterprise corpora.

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